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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

CS2001-1 – DATA STRUCTURES

(For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Why do we write (int *) before malloc? int *ip = (int *)malloc(sizeof(int));
A) It is for the syntax correctness
B) It is for the type-casting
C) It is to inform malloc function about the data-type expected
D) To convert it to a pointer variable
2. How are multidimensional arrays stored in memory?
A) Row-major order
B) Column-major order
C) Both a and b (depending on language)
D) None of the above
3. Which of the following accesses the member x in the structure Point pointed to by ptr?
struct Point { int x; int y; };
struct Point *ptr;
A) ptr->x
B) (*ptr).x
C) Both a and b
D) ptr.x
4. What will be the result of the following operation?
int arr[5] = {1, 2, 3, 4, 5};
printf("%d", *(arr + 2));
A) 1
B) 3
C) 2
D) Compilation error
5. Consider the following stack operations: push(5), push(10), push(20), pop(), push(15), pop(). What will be the final state of the stack?
A) [5, 10]
B) [5, 10, 15]
C) [5, 15]
D) [10, 15]
6. The prefix form of A-B/ (C * D ^ E) is?
A) -A/B*C^DE
B) -A/BC*^DE
C) -ABCD*^DE
D) -/^ACBDE
7. Evaluate the postfix expression 6 2/ 3- 42*+
A) 6
B) 7
C) 8
D) 9
8.

```
int main() { int n,i;
n=f(6); printf("%d",n); }
f(int x) { if (x==2) return 2;
           else { printf("+"); f(x-1); } }
```


A) 2
B) 2+++++
C) +++++2
D) ++++2

9. A circular queue is implemented using an array of size 10. The array index starts with 0, front is 6, and rear is 9. The insertion of next element takes place at the array index.....
 A) 0 B) 7
 C) 9 D) 10
10. In a linear queue of size n , if we perform n insertions followed by n deletions, and then attempt another insertion, what happens?
 A) Insertion is successful B) Queue overflows
 C) Queue underflows D) It depends on the data
11. If a circular queue has front=2 and rear=4 and size=5, how many elements are present if initial value of front = 0 rear = -1
 A) 2 B) 3
 C) 4 D) 5
12. Suppose we insert the elements 15, 10, 30, 5 into an ascending priority queue. The order of deletion will be:
 A) 5, 10, 15, 30 B) 30, 15, 10, 5
 C) 10, 15, 30, 5 D) 15, 30, 10, 5
13. Which is the key disadvantage of using a singly linked list
 A) It require more memory than an array B) It doesn't allow random access of elements
 C) Its difficult to implement D) Its less efficient more large input size
14. When inserting a new node at the beginning of a Circular Singly Linked List, what must be updated?
 A) The last node's link pointer and the new node's link pointer B) Only the new node's link pointer
 C) Only the head pointer D) Only the last node's link pointer
15. Given only a pointer to a node x in a singly linked list (without the head/first pointer), can we delete node x ?
 A) Possible if x is not the last node B) Possible if size of the linked list is even
 C) Possible if size of the linked list is odd D) Possible if x is not the first node
16. Which of the following operations are more efficient in a doubly linked list than in a singly linked list.
 A) Searching for an element B) Traversing the list
 C) Deleting a node when pointer to a node is given. D) Accessing the first element
17. What is the degree of a vertex in a graph?
 A) Number of edges incident to the vertex B) Number of vertices adjacent to it
 C) Number of edges connected to it. D) Number of loops at the vertex
18. Which of the following is NOT a property of a good hashing function?
 A) Deterministic B) Fast computation
 C) Produces the same hash for different inputs D) Uniform distribution
19. What is the maximum number of children that a binary tree node can have?
 A) 0 B) 1
 C) 2 D) 3
20. If the preorder traversal of a binary tree is [A, B, D, E, C, F], and inorder is [D, B, E, A, C, F], the root of the tree is:
 A) A B) B
 C) C D) D

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Define structure and union with a suitable example. Show how to declare and initialize them.

Marks	BT*	CO*	PO*
4	L*2	1	1

- b) Develop a structure to represent the details of employees in a company. Each employee has fields for employee ID, name, and salary.
Write C functions to:
- Read the details of all employees.
 - Print the details of the employee who has the highest salary.
- 8 L3 1 2
- c) Explain how two-dimensional arrays are represented in memory with an example.
- 4 L2 1 1
2. a) Find the output of the following program
- ```
int main() {
 int x = 30, *y, *z, a[4]={10,20,30,40};
 char str[] = "Pointer";
 char *p = str;

 y = &x; z = a;
 x++; p++; z++;
 printf("%s\n", p);
 printf("%d %d %d %d", x, *y, *z, *a);
 return 0;
}
```
- 4 L3 1 2
- b) Write a recursive C program to compute the  $n^{\text{th}}$  Fibonacci number. For  $n = 3$ , Show clearly the stack contents at each stage of recursion and draw the corresponding recursive call tree.
- 8 L3 2 2
- c) Differentiate infix and postfix expression. Write an algorithm to evaluate the postfix expression.
- 4 L2 2 1
3. a) Define stack. Implement push and pop operations for stack using arrays.
- 4 L2 2 1
- b) Write a C function to convert an infix expression to a postfix expression using stack. Explain the working of the function step by step with the help of the example:  $a * (b + c * d)$
- 6 L3 2 1
- c) Write a recursive program to implement binary search.
- 4 L3 2 1
- Unit – II**
4. a) With a suitable example, explain how a linear queue can be implemented using an array.
- 4 L2 3 1
- b) Explain the disadvantage of a linear queue and how it is overcome in a circular queue. Write C functions for the following operations on a circular queue.
- Insert
  - Delete
  - Display
- 8 L3 3 1
- c) What is a linked list? Explain different types of linked lists with neat diagram.
- 4 L2 4 1
5. a) A circular queue of size 5 (indices 0–4) is initially empty (front = -1, rear = -1). Perform the following operations:  
insert(10), insert(20), insert(30), delete(), insert(40), insert(50), delete(), insert(60)  
Draw the circular queue after each operation and indicate the values of front and rear
- 4 L3 3 2
- b) Explain how a stack can be implemented using a singly linked list. Write C functions to implement a stack using a singly linked list. Include functions for push, pop, and display operations.
- 8 L3 4 1

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- |       |                                                                                                                                                               |   |    |   |   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| c)    | Write a C function to insert a node at the beginning of a circular linked list.                                                                               | 4 | L3 | 4 | 1 |
| 6. a) | Write a C function to delete a node with a given key from a singly linked list.                                                                               | 4 | L3 | 4 | 1 |
| b)    | Write C functions for the following operations on a Doubly Linked List:<br>i. Insert a node after a given node<br>ii. Traverse the list in backward direction | 8 | L3 | 4 | 2 |
| c)    | Write a C function to count the number of nodes in a singly linked list.                                                                                      | 4 | L3 | 4 | 2 |

**Unit – III**

- |       |                                                                                                                                                                   |   |    |   |   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 7. a) | Explain in detail the different types of threaded binary trees with diagrams.                                                                                     | 4 | L2 | 5 | 1 |
| b)    | Define Binary Search Tree (BST). Construct a BST for the sequence 45, 12, 54, 9, 20, 47, 70, 16, 28, 68, 48 and show preorder, inorder, and postorder traversals. | 8 | L3 | 5 | 2 |
| c)    | What is Collision? What are the methods to resolve collision.                                                                                                     | 4 | L2 | 5 | 1 |
| 8. a) | Write a function to search for a given key in a Binary Search Tree                                                                                                | 4 | L3 | 5 | 2 |
| b)    | Define a graph. Explain different ways of representing a graph in memory with examples. Compare their advantages and disadvantages.                               | 8 | L2 | 5 | 1 |
| c)    | Write a function to insert a node into a Binary Search Tree.                                                                                                      | 4 | L3 | 5 | 2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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